Biotechnology and Its Applications

Totipotency & Plant Tissue Culture

Totipotency: The capacity to generate a whole plant from a part or tissue of the plant. This process is called tissue culture.

Micro-propagation: The method of producing thousands of plants by a single cell.

Somaclones: The new plants will be a copy of the older one, thus called somaclones.

What if a plant is infected with viruses?

The apical/axillary meristem remains virus-free. Thus it can be grown to produce a healthy plant.

e.g., banana, sugarcane, potato

Somatic Hybridisation: When protoplasts from 2 plants are isolated and fused to get hybrid protoplasts → can be grown further.

GMOs (Genetically Modified Organisms): Organisms that are genetically modified. Used to alter features for food (like starch), fuel, pharma, etc.

Bt Toxin & Pest Resistance

Origin: Produced from *Bacillus thuringiensis*. Cloned and induced in plants to make them pest-resistant (e.g., Bt cotton, Bt corn, Bt tomato).

Target Insects

- Lepidopterans: tobacco budworms, armyworms
- Coleopterans: beetles
- Dipterans: flies, mosquitos

Bt cotton: Some strains of Bacillus thuringiensis prod proteins that kill bactract like Lepid-Coleopterans, Coleop, Dipterans

Bt forms protein crystals during a phase of their growth. It kills insects due to the presence of a toxic insecticidal protein in the crystals.

Genetic Coding of Bt & The Nematode Threat

cryIAc & cryIIAb	controls cotton bollworms
cryIAb	controls corn borer

Nematode Threat!

The nematode Meloidogyne incognita infects the roots of tobacco plants. This causes a severe reduction in yields.

The RNA Interference (RNAi) Mechanism

Agrobacterium vectors are used to introduce nematode-specific genes into specific plants.

↓

The plant produces both sense and anti-sense RNA (which are complementary to each other).

↓

This initiates RNAi

↓

Silences specific mRNA due to complementary dsRNA that binds to and prevents translation of the mRNA.

↓

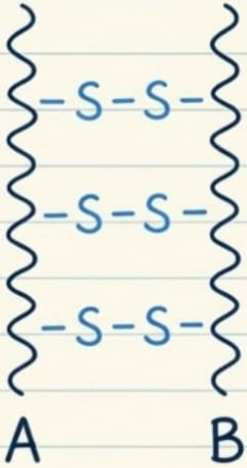
Result: The parasite could not survive, and thus the plant gets protected!

Insulin Production & Structure

The Old Way:
Previously extracted from the glands of slaughtered pigs and cows.

But this caused allergies in patients!

Pro-hormone Processing:
Synthesised as a pro-hormone containing an extra stretch called the C-peptide. It gets processed so mature insulin is formed and the C-peptide is removed.



It consists of two short polypeptide chains linked together by disulfide bonds.

1983: The Eli Lilly Breakthrough

In 1983, Eli Lilly (an American company) prepared two DNA sequences corresponding to the A and B chains of human insulin.

Introduced them into plasmids of *E. coli* to produce insulin chains. Chains A and B were produced separately.

They were extracted and combined by creating disulfide bonds to form perfectly synthesised human insulin.

First synthetic human insulin — solved the allergy problem!

Gene Therapy & ADA Deficiency

Definition: A collection of methods allowing the correction of a gene defect diagnosed in a child/embryo by delivery of a normal gene.

Case Study (1990): First clinical gene therapy given for Adenosine deaminase (ADA) deficiency. ADA is crucial for the immune system to function. function. (Caused by deletion of the gene).

Current Non-Curative Treatments: Bone marrow transplantation or enzyme replacement therapy.

The Gene Therapy Process

1. Lymphocytes from the patient's blood are grown outside the body.
2. A functional ADA cDNA is introduced into these lymphocytes using a retroviral vector.
3. The engineered cells are returned to the patient.

Because these cells are not immortal, the patient needs periodic infusion! A permanent cure requires introduction at early embryonic stages.

Molecular Diagnosis

Context: Early understanding of pathophysiology is critical.
Traditional serum and urine analysis cannot provide early detection.

PCR (Polymerase Chain Reaction)

- Detects very low concentrations of bacteria/viruses by amplification of their
- Used to detect HIV in suspected AIDS patients and mutations in genes in suspected cancer patients.

Probes & Autoradiography

- A single-stranded DNA/RNA tagged with a radioactive molecule (a probe).
- Allowed to hybridise to its complementary DNA in a clone of cells.
- Followed by detection using autoradiography.

Result: The mutated gene will not appear on the photographic film, because the probe lacks complementarity with the mutated gene.

Transgenic Animals: Research Applications

Definition: Animals that have had their DNA manipulated to possess and express an extra (foreign) gene. Over 95% of all existing **transgenic animals** are mice.

Normal Physiology and Development

- Designed to allow the study of how genes are regulated and affect normal body functions.
- Example: Studying complex factors involved in growth, such as insulin-like growth factor.

Study of Disease

- Serve as models for human diseases so that investigation of new treatments is possible.
- Today, models exist for many diseases: cancer, cystic fibrosis, rheumatoid arthritis, and Alzheimer's.

Transgenic Animals: Products & Safety

Biological Products

Used to produce biological products that are otherwise expensive to make (e.g., human protein α-1-antitrypsin used to treat emphysema).

1997: The first transgenic cow, Rosie, produced human protein-enriched milk (2.4 grams per litre, containing human alpha-lactalbumin). More nutritionally balanced for human babies.

Vaccine Safety

Transgenic mice are being developed for use in testing the safety of vaccines (like the polio vaccine). If reliable, they could replace the use of monkeys.

Chemical Safety Testing

Also known as toxicity/safety testing. Animals carry genes making them more sensitive to toxic substances. Allows us to obtain toxicity results in less time.

Ethical Issues, Biopiracy & Regulation

The Ecosystem Problem: Genetic modification of organisms can have unpredictable results when introduced into the ecosystem. Regulation is mandatory.

GEAC (Genetic Engineering Approval Committee): Set up by the Indian Government to make decisions regarding the validity of GM research and the safety of introducing GM-organisms for public services.

Industrialised nations = rich financially, poor in biodiversity. Developing world = rich in biodiversity & traditional knowledge.

Biopiracy: The use of bio-resources by multinational companies and other organisations without proper authorisation from the countries and people concerned, without compensatory payment.

The Legal Solution

The Indian Parliament has cleared the second amendment of the Indian Patents Bill, which takes such issues into consideration, including patent terms emergency provisions and research and development initiatives.

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